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Monocular underwater camera biomass estimation

Abstract

Methods, systems, and apparatus, including computer programs encoded on computer-storage media, for monocular underwater camera biomass estimation. In some implementations, an exemplary method includes obtaining a plurality of images of fish captured by a monocular underwater camera; providing the plurality of images that were captured by the monocular underwater camera to a first model trained to detect one or more fish within the plurality of images; generating one or more values for each detected fish as a set of values; generating a biomass distribution of the fish based on the set of values; and determining an action based on the biomass distribution.

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Background/Summary

FIELD

(1) This specification generally relates to cameras that are used for biomass estimation, and particularly to underwater cameras that are used for aquatic livestock.

BACKGROUND

(2) A population of farmed fish may include fish of varying sizes, shapes, and health conditions. In the aquaculture context, prior to harvesting, a worker may remove some fish from the fish pen and weigh them. The manual process of removing the fish from the fish pen and weighing them is both time intensive and potentially harmful to the fish. In addition, because only a small portion of a fish population may be effectively measured in this way, the true characteristics of the population remain unknown.

SUMMARY

(3) In general, innovative aspects of the subject matter described in this specification relate to estimating the biomass of aquatic livestock using monocular underwater camera systems. Individual fish can be photographed using a single underwater camera. Images from the underwater camera can be processed using computer vision and machine learning-based techniques to identify fish within the images and to determine features, e.g., truss lengths, on the fish. Biomass estimations are generated by a model (e.g., neural network, Random Forest Regressor, Support Vector Regressor, or Gaussian Process Regressor, among others) that is trained to generate predicted biomass based, e.g., on truss lengths. The biomass of fish populations may be used to control the amount of feed given to a fish population, e.g., by controlling a feed distribution system, as well as to identify and isolate runt, diseased, or other sub-populations.

(4) One innovative aspect of the subject matter described in this specification is embodied in a method that includes obtaining a plurality of images of fish captured by a monocular underwater camera; providing the plurality of images that were captured by the monocular underwater camera

to a first model trained to detect one or more fish within the plurality of images; generating one or more values for each detected fish as a set of values; generating a biomass distribution of the fish based on the set of values; and determining an action based on the biomass distribution.

(5) Other implementations of this and other aspects include corresponding systems, apparatus, and computer programs, configured to perform the actions of the methods, encoded on computer storage devices. A system of one or more computers can be so configured by virtue of software, firmware, hardware, or a combination of them installed on the system that in operation cause the system to perform the actions. One or more computer programs can be so configured by virtue of having instructions that, when executed by data processing apparatus, cause the apparatus to perform the actions.

(6) The foregoing and other embodiments can each optionally include one or more of the following features, alone or in combination. For instance, in some implementations, generating the biomass distribution of the fish based on the set of values includes providing the set of values to a second model trained to estimate biomass distributions; and obtaining output of the second model as the biomass distribution of the fish.

(7) In some implementations, the one or more values for each detected fish include a value that indicates a length between a first point on a particular fish of the fish and a second point on the particular fish.

(8) In some implementations, actions include detecting the first point and second point on the particular fish. In some implementations, the action includes adjusting a feeding system providing feed to the fish.

(9) In some implementations, the action includes sending data including the biomass estimate to a user device, where the data is configured to, when displayed on the user device, present a user of the user device with a visual representation of the biomass estimate.

(10) In some implementations, actions include obtaining data from the monocular underwater camera indicating a current operation status of the monocular underwater camera; and in response to obtaining the plurality of images from the monocular underwater camera, providing the plurality of images to the first model.

(11) An advantage of the methods, systems, and apparatuses described herein includes reducing reliance on hardware for obtaining depth data from an environment such as hardware for stereo cameras, light based depth detection, among others, e.g., by using processes to generate accurate biomass estimations from images captured by monocular or single lens cameras. When a setup of stereo cameras is used, depth information can be determined from the visual differences of objects captured by the two or more cameras of the stereo camera setup. However, if a stereo camera setup is damaged, is partially obscured, or is partially made non-functional, processes that rely on depth information of the stereo camera may not function properly. Similarly, stereo camera setups generally are more expensive to produce, calibrate, and maintain than single camera equivalents. Stereo camera setups can also be less efficient by requiring greater amounts of image data to be transferred to processing elements compared to single camera equivalents.

(12) Solutions described herein specify the use of two-dimensional (2d) truss networks to determine biomass distributions for a population. Typically, truss networks are computed in three dimensions to account for changes in apparent distances between key points based on a position of a fish or other object in three dimensions. However, this approach relies on hardware for obtaining depth data from an environment such as hardware for stereo cameras, light based depth detection, among others. Solutions described herein specify the use of a model trained with 2d truss networks to generate an accurate biomass distribution prediction for a population of fish.

(13) Other possible solutions, such as a time of flight (ToF) sensor present further issues, including environmental effects, that can affect accuracy. For example, ToF sensors or other depth sensors can detect debris within an environment, such as water, as objects for distance measurement. The variable debris in the environment can similarly present issues in determining a correlation of time

for reflection and actual distance as the time may depend on the amount of debris in the environment.

(14) The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features and advantages of the invention will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram showing an example of a system that is used for monocular underwater camera biomass estimation.

(2) FIG. 2 is a flow diagram showing an example of a system for monocular underwater camera biomass estimation.

(3) FIG. 3 is a diagram showing an example of a truss network.

(4) FIG. 4 is a diagram illustrating an example of a computing system used for monocular underwater camera biomass estimation.

(5) Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

(6) FIG. 1 is a diagram showing an example of a system **100** that is used for underwater camera biomass estimation. The system **100** includes a control unit **116** and an underwater camera device **102**. Generally speaking, the control unit **116** obtains images captured by a camera of the camera device **102** and processes the images to generate biomass estimations for one or more fish. The biomass estimations for one or more fish can be processed to determine actions such as feed adjustment, sorting, model training, and user report feedback, among others.

(7) In some implementations, the control unit **116** includes one or more components for processing data. For example, the control unit **116** can include an object detector **118**, truss length engine **120**, biomass engine **124**, and a decision engine **128**. Components can include one or more processes that are executed by the control unit **116**.

(8) At least one of the one or more cameras of the camera device **102** includes a camera that captures images from a single viewpoint at a time. This type of camera may be referred to as a monocular camera. Where a stereo camera setup can include multiple cameras each capturing a unique viewpoint at a particular time, a monocular camera captures one viewpoint at a particular time. A computer processing output of a stereo camera setup can determine, based on differences in the appearance of objects in one viewpoint compared to another viewpoint at a particular time, depth information of the objects.

(9) In some implementations, the camera device **102** has one or more cameras in a stereo camera setup that are non-functional or obscured, e.g., by debris or other objects, including fish or other animals, in an environment. In some implementations, the control unit **116** can process one or more images from the camera device **102**, or obtain a signal from the camera device **102**, indicating that one or more cameras of the camera device **102** are obscured or non-functional or the camera device **102** is operating with a monocular camera. The control unit **116** can adjust a processing method based on the status of the cameras of the camera device **102**, such as a stereo setup status or a monocular camera setup status. In some implementations, a monocular underwater camera includes stereo camera setups that have become, operationally, a monocular camera setup based on non-functioning elements, debris, among other causes. In some cases, stereo camera setups can obtain images from a single viewpoint at a given time and be considered, operationally, a monocular camera.

(10) In some implementations, the camera device **102** includes a single camera. For example, the single camera can be a monocular camera that captures a single viewpoint at a time. In some

implementations, the camera device **102** with a single camera is more efficient to produce and maintain, more economical, and can be more robust with fewer components prone to failure.

(11) The system **100** also includes a feed controller unit **130** that controls the feed delivered by feed system **132**. The feed controller unit **130** can include components configured to send control messages to actuators, blowers, conveyers, switches, or other components of the feed system **132**. The control messages can be configured to stop, start, or change a meal provided to fish **106** in pen **104**.

(12) In this example, the camera device **102** includes propellers to move the camera device **102** around the fish pen **104**. In general, the camera device **102** may use any method of movement including ropes and winches, waterjets, thrusters, tethers, buoyancy control apparatus, chains, among others.

(13) In some implementations, the camera device **102** is equipped with the control unit **116** as an onboard component, while in other implementations, the control unit **116** is not affixed to the camera device **102** and is external to the camera device **102**. For example, the camera device **102** may provide images **112** and **114** over a network to the control unit **116**. Similarly, the control unit **116** can provide return data, including movement commands to the camera device **102** over the network.

(14) Stages A through C of FIG. 1, depict image data **110**, including image **112**, obtained by the camera device **102** that are processed by the control unit **116**. The image **112** includes representations of the fish **113** and **115**. Images of fish obtained by the camera device **102** may include fish in any conceivable pose including head on, reverse head on, or skewed.

(15) In stage A, the camera device **102** obtains the image data **110** including image **112** of the fish **113** and **115** within the pen **104**. The camera device **102** provides the data **110** to the control unit **116**. In some implementations, the camera device **102** obtains multiple images and provides the multiple images, including the image **112**, to the control unit **116**.

(16) In stage B, the control unit **116** processes the images of the data **110**, including the image **112**. The control unit **116** provides the data **110** to the object detector **118**. The object detector **118** can run on the control unit **116** or be communicably connected to the control unit **116**. The object detector **118** detects one or more objects in the images of the data **110**. The one or more objects can include large scale objects, such as fish, as well as smaller objects, such as key points of the fish.

(17) In the example of FIG. 1, the object detector **118** detects the fish **113** and the fish **115** in the image **112**. The object detector **118** can compute a bounding box for each detection. For example, the object detector **118** can compute a bounding box **118a** indicating a location of the fish **113** and a bounding box **118b** indicating a location of the fish **115**. The object detector **118** can compute a bounding boxes for key points **118c**. Instead of bounding boxes, other types of data indicating a location can be used, such as a specific location specified by one or more values (e.g., x and y values in a coordinate plane of the image **112**, spherical coordinates, among others).

(18) The object detector **118** detects key points **118c** of the fish **113** and fish **115**. As shown in FIG. 1, the key points **118c** of the fish **113** and the fish **115** represent locations on the fish **113** and the fish **115**. The locations can include a mouth, upper fin, tail, eye, among others. The object detector **118** can include one or more machine learning models. The machine learning models can be trained to detect key points as well as objects such as fish. In some implementations, the machine learning models are trained by the control unit **116**.

(19) In some implementations, machine learning models of the object detector **118** are trained to detect key points. For example, the object detector **118** can obtain images including one or more key points. The object detector **118** can determine a predicted location for each of the one or more key points. The object detector **118** can obtain ground truth data indicating an actual location for each of the one or more key points. Based on comparing the ground truth data to the predicted location for the one or more key points, the object detector **118** can determine an error term. The object detector **118** can then adjust one or more parameters of the machine learning models

according to the error term in order to improve subsequent predictions.

(20) In some implementations, machine learning models of the object detector **118** are trained to detect fish or other animals. For example, the object detector **118** can obtain images including one or more depictions of fish. The object detector **118** can determine a predicted location for the one or more depictions of fish. The object detector **118** can obtain ground truth data indicating an actual location for the depicted fish. Based on comparing the ground truth data to the predicted location for the depicted fish, the object detector **118** can determine an error term. The object detector **118** can then adjust one or more parameters of the machine learning models according to the error term in order to improve subsequent predictions.

(21) The truss network engine **114** obtains detected objects from the object detector **118**. In some implementations, the control unit **116** obtains detected object data from the object detector **118** and provides the detected object data to the truss network engine **114**. The truss network engine **114** generates truss networks **120a** and **120b** for the fish **113** and the fish **115**, respectively. The truss network **120b** for the fish **115** is shown in more detail in FIG. 3. In general, the truss networks **120a** and **120b** indicate distances between locations detected on the fish **113** and **115**. These distances can be associated with unique properties of the fish such as biomass, identity, or health.

(22) In some implementations, the truss networks **120a** and **120b** are two dimensional (2d) truss networks. For example, the truss network engine **114** determines a distance between a first key point of the key points **118c** and a second key point of the key points **118c** based on a number of pixels in the image **112** separating the first key point from the second key point. Such a determination does not rely on depth information from the image **112**. Truss lengths generated in such a way may inaccurately indicate features of a fish, such as biomass, based on a pose of the fish (e.g., truss lengths for fish faced head on towards the camera **102** can appear smaller than they would if the fish was faced side on). However, by processing generated truss networks in aggregate, the biomass engine **124** can generate accurate predictions for the population of fish **106** as historical training data likely has a similar distribution of fish poses and corresponding generated truss networks. The control unit **116** can generate accurate biomass distributions for the fish **106**.

(23) The truss network engine **114** generates truss data **122** including the truss networks **120a** and **120b**. In some implementations, the truss data **122** includes truss networks generated for multiple fish, such as all the fish **106**. The biomass engine **124** obtains the truss data **122**. In some implementations, the control unit **116** obtains the truss data **122**, including the truss networks **120a** and **120b**, and provides the truss data to the biomass engine **124**.

(24) The biomass engine **124** generates a biomass distribution **126** based on the truss data **122**. In some implementations, the biomass engine **124** includes one or more machine learning models trained to generate biomass distributions for one or more fish. For example, the biomass engine **124** can obtain truss networks for a population of fish. The biomass engine **124**, and one or more models therein, can generate a predicted biomass distribution for the population of fish based on the obtained truss networks. The biomass engine **124** can then compare the predicted biomass distribution with a known, or previously calculated, biomass distribution for the population of fish. The biomass engine **124** can adjust one or more parameters of the one or more machine learning models of the biomass engine **124** based on a result of comparing the predicted biomass distribution to the known biomass distribution. Various training techniques, including gradient descent, backpropagation, among others, can be used in training models of components of the control unit **116**, including the biomass engine **124** component.

(25) The known biomass distribution can be determined using one or more algorithms, models, or manual weighing of fish. The population of fish from which training data is derived can be different from the fish **106**. The population of fish can be stored in a historical database including records from previous populations of fish. In some implementations, the control unit **116**, or processors configured to operate similar to the control unit **116**, generates truss data and provides the truss data to a database for later use in training. For example, the control unit **116** can provide

the truss data **122** to a database. The truss data **122** can be stored and used for later processes, such as subsequent training of one or more models. In general, training data can include 2d truss networks as described herein.

(26) The biomass distribution **126** generated by the biomass engine **124** includes one or more biomasses, corresponding to one or more fish, of the fish **106**. In the example of FIG. **1**, a number of fish corresponding to ranges of biomass is determined by the biomass engine **124**. For example, the biomass engine **124** can determine that two fish correspond to biomass range 3.5 kilogram to 3.7 kilogram and three fish correspond to the biomass range 3.7 kilogram to 3.8 kilogram. In some implementations, the ranges are predetermined. In some implementations, the ranges are dynamically chosen by the control unit **116** based on the number of fish and the distribution of biomasses. In some implementations, the biomass distribution **126** is a histogram.

(27) The biomass engine **124** generates the biomass distribution **126** and provides the biomass distribution **126** to the decision engine **128**. The decision engine **128** obtains the biomass distribution **126**. In stage C, the control unit **116** determines an action based on the biomass distribution **126**. In some implementations, the control unit **116** provides the biomass distribution **126**, including one or more biomass estimation values, to the decision engine **128**.

(28) In some implementations, the biomass engine **124** generates the biomass distribution **126** that includes likelihoods that a number of fish of the fish **106** are a particular biomass or within a range of biomass. For example, weight ranges for the biomass distribution **126** can include ranges from 3 to 3.1 kilograms (kg), 3.1 to 3.2 kg, and 3.2 to 3.3 kg. A likelihood that a number of the fish **106** are within the first range, 3 to 3.1 kg, can be 10 percent. A likelihood that the number of the fish **106** are within the second or third range, 3.1 to 3.2 kg or 3.2 to 3.3 kg, respectively, can be 15 percent and 13 percent. In general, the sum of all likelihoods across all weight ranges can be normalized (e.g., equal to a value, such as 1, or percent such as 100 percent).

(29) In some implementations, the decision engine **128** determines a portion of the fish **106**, based on data generated by the biomass engine **124**, are below an expected weight or below weights of others of the fish **106**. For example, the decision engine **128** can determine subpopulations within the fish **106** and determine one or more actions based on the determined subpopulations, such as actions to mitigate or correct for issues (e.g., runtting, health issues, infections, disfiguration, among others). Actions can include feed adjustment, sorting, model training, and user report feedback, among others.

(30) In some implementations, the decision engine **128** detects one or more features of the biomass estimations of the biomass distribution **126**. For example, the decision engine **128** can detect one or more subpopulations. Subpopulations can include runt fish, healthy fish, diseased fish, among others. For example, a Gaussian-like shape in the biomass distribution **126** can indicate a subpopulation. In the example of FIG. **1**, the biomass distribution **126** includes at least two subpopulations with one having a smaller average biomass than the other. The control unit **116** can determine the subpopulation with the smaller average biomass is a runt population based on a comparison with runt population criteria, such as a runt population average biomass threshold, among others.

(31) In some implementations, the decision engine **128** detects a runt subpopulation based on processing the biomass distribution **126**. For example, the decision engine **128** can include one or more algorithms or trained models to detect groups within a distribution of data. The decision engine **128** can include one or more processors configured to perform clustering algorithms such as k-mean, partitioning methods, hierarchical clustering, fuzzy clustering, density-based clustering, model-based clustering, among others.

(32) In some implementations, the control unit **116** determines an adjustment of feed using the feed controller unit **130** controlling the feed system **132**. The control unit **116** can provide the biomass distribution **126** or a control signal to the feed controller unit **130**. Depending on the data received from the control unit **116**, the feed controller unit **130** can either process the biomass distribution

126 to determine an adjustment of feed and provide a control signal to the feed system **132** or can provide the control signal provided by the control unit **116** to the feed system **132**.

(33) In some implementations, the decision engine **128** does not detect a runt subpopulation. For example, the decision engine **128** can determine that the biomass distribution **126** does or does not satisfy a biomass requirement or threshold, such as a biomass requirement for distribution or sale. The decision engine **128** can determine, based on features of the biomass distribution **126**, what action to perform.

(34) For example, if one or more biomass estimations generated by the biomass engine **124** do not satisfy a threshold (e.g., the mean or median biomass is too large or too small), the control unit **116** can provide a control signal to a sorting actuator of a sorting system to sort one or more fish from the fish **106** or can provide a control signal to adjust a feeding of the fish **106**. For example, the control unit **116** can sort the fish **106** based on biomass. The control unit **116** can send a signal to a sorting system that sorts fish based on one or more criteria, such as a threshold biomass, into multiple locations based on the one or more criteria.

(35) In some implementations, the control unit **116** includes the feed controller unit **130**. For example, the control unit **116** may control both the processing of the images in the data **110** and the adjustments to the feeding by controlling the feed system **132**.

(36) In some implementations, the control unit **116** adjusts feeding to provide feed to a certain area of the fish pen **104**. For example, the obtained data **110** can include positions of the fish detected within the images of the obtained data **110**. The control unit **116** can determine based on one or more subpopulations detected by the decision engine **128** of the control unit **116** that a given subpopulation requires additional feed.

(37) The control unit **116** can send a control signal to the feed system **132** or to the control unit **130** for the feed system **132** configured to adjust the location of an output of feed. The control unit **116** can adjust the location of an output of feed to a location of one or more fish within a particular subpopulation or an average location of the subpopulation.

(38) In some implementations, the feed system **132** includes multiple food types. For example, the controller unit **130** can provide control messages to the feed system **132** to change the food type provided to the fish **106**. In some cases, the multiple food types include a medicated food type and a non-medicated food type. In some cases, the multiple food types include food with a particular nutritional value and food with a different nutritional value.

(39) The controller unit **130** can determine, based on data from the control unit **116**, which food to provide to the fish **106**, how much food to provide, when to provide the food, and at what rate to provide the food. In general, the controller unit **130** can generate a meal plan based on data from the control unit **116**, such as biomass estimations or a control signal generated based on biomass estimations, where the meal plan includes one or more of: a feed type, a feed rate, a feed time, and a feed amount.

(40) In some implementations, the control unit **116** includes multiple computer processors. For example, the control unit **116** can include a first and a second computer processor communicably connected to one another. The first and the second computer processor can be connected by a wired or wireless connection. The first computer processor can perform one or more of the operations of the object detector **118**, truss length engine **120**, biomass engine **124**, or the decision engine **128**. The first computer processor can store or provide training data to train one or more models of the object detector **118**, truss length engine **120**, biomass engine **124**, or the decision engine **128**.

(41) Similarly, the second computer processor can perform one or more of the operations of the object detector **118**, truss length engine **120**, biomass engine **124**, or the decision engine **128**. The second computer processor can store or provide the training data to train one or more models of the object detector **118**, truss length engine **120**, biomass engine **124**, or the decision engine **128**.

Operations not performed by the first computer processor can be performed by the second computer processor or an additional computer processor. Operations not performed by the second

computer processor can be performed by the first computer processor or an additional computer processor.

(42) In some implementations, the control unit **116** operates one or more processing components, such as the object detector **118**, truss length engine **120**, biomass engine **124**, or the decision engine **128**. In some implementations, the control unit **116** communicates with an external processor that operates one or more of the processing components. The control unit **116** can store training data, or other data used to train one or more models of the processing components, or can communicate with an external storage device that stores data including training data.

(43) In some implementations, models of the components of the control unit **116** include one or more fully or partially connected layers. Each of the layers can include one or more parameter values indicating an output of the layers. The layers of the models can generate output for each of the components, such as the object detector **118**, truss length engine **120**, biomass engine **124**, or the decision engine **128**.

(44) In general, the control unit **116** can process one or more images of the data **110** in aggregate or process each image of the data **110** individually. In some implementations, one or more components of the control unit **116** process items of the data individually and one or more components of the control unit **116** process the data **110** in aggregate. For example, the object detector **118** can process each image of the data **110** individually. Similarly, the truss length engine **120** can obtain data from the object detector **118** and process each truss network for each detected fish. In some implementations, parallel processes of the object detector **118** and the truss length engine **120** generate object detections and truss lengths corresponding to the object detections for multiple fish at a time. For example, a first process of detecting a fish object and then determining a truss network for that fish can proceed as a parallel process with a second process of detecting another fish object and then determining a truss network for that fish.

(45) In general, processing individually as discussed herein, can include processing one or more items sequentially or in parallel using one or more processors. In some implementations, the decision engine **128** processes data in aggregate. For example, the decision engine **128** can determine, based on one or more data values indicating biomass estimates provided by the biomass engine **124**, one or more decisions and related actions, as described herein. In general, processing in aggregate as discussed herein, can include processing data corresponding to two or more items of the data **110** to generate a single result. In some implementations, an item of the data **110** includes the image **112**.

(46) FIG. 2 is a flow diagram showing an example of a process **200** for monocular underwater camera biomass estimation. The process **200** may be performed by one or more systems, for example, the system **100** of FIG. 1.

(47) The process **200** includes obtaining images captured by a monocular underwater camera (**202**). For example, the control unit **116** obtains images of the image data **110** captured by the camera device **102**. The camera device **102** can include a single camera. The camera device **102** with a single camera can be more efficient to produce and maintain, more economical, and can be more robust with fewer components prone to failure.

(48) The process **200** includes providing data corresponding to multiple fish based on the images to a trained model (**204**). For example, the control unit **116** provides the image data **110** to the object detector **118** trained to detect objects in images of the image data **110**. In another example, the control unit **116** provides the image data **110** to the object detector **118** and the truss length engine **120** to generate the truss data **122**. The control unit **116** then provides the truss data **122** to the biomass engine **124**. The biomass engine **124** includes one or more models trained to determine one or more biomass estimations based on one or more input truss networks.

(49) The process **200** includes obtaining output of the trained model including one or more biomass values (**206**). For example, the control unit **116** can obtain output of the models, including the object detector **118** and the biomass engine **124**. The output of the object detector **118** can include

detected objects such as fish or key points. The output of the biomass engine **124** can include biomass distribution **126**.

(50) The process **200** includes determining an action based on the one or more biomass values (**208**). For example, the control unit **116** can determine to adjust a feeding of the fish pen **104** based on the detected objects and the biomass distribution **126**. The decision engine **128** component of the control unit **116** can choose from one or more available actions, e.g., available actions obtained from data or provided by a user. In the example of FIG. **1**, the available actions include feed adjustment. The control unit **116** can provide a signal to the feed controller unit **130** to adjust the feed for the fish pen **104**.

(51) In some implementations, the action includes sending data including the biomass distribution **126** to a user device, where the data is configured to, when displayed on the user device, present a user of the user device with a visual representation of the biomass distribution **126**. For example, the control unit **116** can generate a data signal that includes an indication of one or more biomass estimates, such as the biomass distribution **126**. In some implementations, the control unit **116** waits for feedback from a user provided a visual representation of a biomass estimate to confirm an action determined by the decision engine **128**, such as a feed adjustment.

(52) In some implementations, the control unit **116** obtains data from a monocular underwater camera indicating a current operation status of the monocular underwater camera and, in response to obtaining an image of a fish, provides the image of the fish to a trained model. For example, the camera **102** may include a dysfunctional stereo camera pair. The dysfunction can result in images without the depth data provided by the stereo effect of the cameras. To mitigate this situation, the camera device **102** can send a signal to the control unit **116** indicating a camera of a stereo pair is dysfunctional after the camera device **102** determines a camera of a stereo pair has become dysfunctional. In response to the signal, the control unit **116** can process images obtained from the camera device **102** as discussed herein.

(53) In some implementations, the control unit **116** determines, based on the data **110** or other data provided by the camera device **102**, the images from the camera device **102** do not include depth data. For example, the control unit **116** can process the data **110** and determine that images lack a stereo feature. In response, the control unit **116** can process the images to determine a biomass distribution without this additional depth data. For example, as discussed herein, the control unit **116** can generate a group of truss networks for a population of fish, such as the fish **106**, and provide the group of truss networks to the biomass engine **124** to determine a biomass distribution for the fish **106**.

(54) The processing switch can be automatic such that, intermediate data processing issues or environmental effects, such as fish or debris obfuscating or blocking an image of a stereo pair, can be detected by the control unit **116** and the control unit **116** can determine to process corresponding images as described herein.

(55) FIG. **3** is a diagram showing an example of a truss network **120b**. FIG. **3** shows truss lengths and key points computed for the fish **115** by the system **100** shown in FIG. **1**. The truss lengths between key points are used to extract information about the fish including a weight of the fish. Various trusses, or lengths between key points, of the fish can be used. FIG. **3** shows a number of possible truss lengths including upper lip **302** to eye **304**, upper lip **302** to leading edge dorsal fin **306**, upper lip **302** to leading edge pectoral fin **308**, leading edge dorsal fin **306** to leading edge anal fin **310**, leading edge anal fin **310** to trailing low caudal peduncle **312**, trailing lower caudal peduncle **312** to trailing upper caudal peduncle **314**. Other key points and other separations, including permutations of key points mentioned, can be used. For different fish, or different species of fish, different key points may be generated. For any set of key points, a truss network may be generated as a model.

(56) Other truss lengths not shown can be used by the system **100**. For example, a truss length from the upper lip **302** to the tail **313** can be used as the length of the fish **115** and included in a

collection of one or more truss length measurements and provided to the biomass engine **124** to generate a biomass distribution. In addition, specific truss lengths can be used to recognize specific deformities. Deformities such as shortened operculum can be detected using truss lengths such as a truss length from the upper lip **302** to the gill. Processing one or more images of a fish can include determining any of the following health conditions: shortened abdomen, shortened tail, scoliosis, lordosis, kyphosis, deformed upper jaw, deformed lower jaw, shortened operculum, runting or cardiomyopathy syndrome (CMS).

(57) In some implementations, a biomass distribution, such as the biomass distribution **126**, includes health data related to one or more fish represented in a distribution. For example, the biomass engine **124** can generate a distribution of one or more fish that includes health probabilities as well as, or instead of, biomass data. The health probabilities can be used to determine various remedial actions including providing medicated feed or moving the fish to a system for treatment, such as delousing.

(58) FIG. **4** is a diagram illustrating an example of a computing system used for monocular underwater camera biomass estimation. The computing system includes computing device **400** and a mobile computing device **450** that can be used to implement the techniques described herein. For example, one or more components of the system **100** could be an example of the computing device **400** or the mobile computing device **450**, such as a computer system implementing the control unit **116**, devices that access information from the control unit **116**, or a server that accesses or stores information regarding the operations performed by the control unit **116**.

(59) The computing device **400** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The mobile computing device **450** is intended to represent various forms of mobile devices, such as personal digital assistants, cellular telephones, smart-phones, mobile embedded radio systems, radio diagnostic computing devices, and other similar computing devices. The components shown here, their connections and relationships, and their functions, are meant to be examples only and are not meant to be limiting.

(60) The computing device **400** includes a processor **402**, a memory **404**, a storage device **406**, a high-speed interface **408** connecting to the memory **404** and multiple high-speed expansion ports **410**, and a low-speed interface **412** connecting to a low-speed expansion port **414** and the storage device **406**. Each of the processor **402**, the memory **404**, the storage device **406**, the high-speed interface **408**, the high-speed expansion ports **410**, and the low-speed interface **412**, are interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **402** can process instructions for execution within the computing device **400**, including instructions stored in the memory **404** or on the storage device **406** to display graphical information for a GUI on an external input/output device, such as a display **416** coupled to the high-speed interface **408**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. In addition, multiple computing devices may be connected, with each device providing portions of the operations (e.g., as a server bank, a group of blade servers, or a multi-processor system). In some implementations, the processor **402** is a single threaded processor. In some implementations, the processor **402** is a multi-threaded processor. In some implementations, the processor **402** is a quantum computer.

(61) The memory **404** stores information within the computing device **400**. In some implementations, the memory **404** is a volatile memory unit or units. In some implementations, the memory **404** is a non-volatile memory unit or units. The memory **404** may also be another form of computer-readable medium, such as a magnetic or optical disk.

(62) The storage device **406** is capable of providing mass storage for the computing device **400**. In some implementations, the storage device **406** may be or include a computer-readable medium, such as a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash

memory or other similar solid-state memory device, or an array of devices, including devices in a storage area network or other configurations. Instructions can be stored in an information carrier. The instructions, when executed by one or more processing devices (for example, processor **402**), perform one or more methods, such as those described above. The instructions can also be stored by one or more storage devices such as computer- or machine readable mediums (for example, the memory **404**, the storage device **406**, or memory on the processor **402**). The high-speed interface **408** manages bandwidth-intensive operations for the computing device **400**, while the low-speed interface **412** manages lower bandwidth-intensive operations. Such allocation of functions is an example only. In some implementations, the high speed interface **408** is coupled to the memory **404**, the display **416** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **410**, which may accept various expansion cards (not shown). In the implementation, the low-speed interface **412** is coupled to the storage device **406** and the low-speed expansion port **414**. The low-speed expansion port **414**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet) may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

(63) The computing device **400** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a standard server **420**, or multiple times in a group of such servers. In addition, it may be implemented in a personal computer such as a laptop computer **422**. It may also be implemented as part of a rack server system **424**. Alternatively, components from the computing device **400** may be combined with other components in a mobile device, such as a mobile computing device **450**. Each of such devices may include one or more of the computing device **400** and the mobile computing device **450**, and an entire system may be made up of multiple computing devices communicating with each other.

(64) The mobile computing device **450** includes a processor **452**, a memory **464**, an input/output device such as a display **454**, a communication interface **466**, and a transceiver **468**, among other components. The mobile computing device **450** may also be provided with a storage device, such as a micro-drive or other device, to provide additional storage. Each of the processor **452**, the memory **464**, the display **454**, the communication interface **466**, and the transceiver **468**, are interconnected using various buses, and several of the components may be mounted on a common motherboard or in other manners as appropriate.

(65) The processor **452** can execute instructions within the mobile computing device **450**, including instructions stored in the memory **464**. The processor **452** may be implemented as a chipset of chips that include separate and multiple analog and digital processors. The processor **452** may provide, for example, for coordination of the other components of the mobile computing device **450**, such as control of user interfaces, applications run by the mobile computing device **450**, and wireless communication by the mobile computing device **450**.

(66) The processor **452** may communicate with a user through a control interface **458** and a display interface **456** coupled to the display **454**. The display **454** may be, for example, a TFT (Thin-Film-Transistor Liquid Crystal Display) display or an OLED (Organic Light Emitting Diode) display, or other appropriate display technology. The display interface **456** may include appropriate circuitry for driving the display **454** to present graphical and other information to a user. The control interface **458** may receive commands from a user and convert them for submission to the processor **452**. In addition, an external interface **462** may provide communication with the processor **452**, so as to enable near area communication of the mobile computing device **450** with other devices. The external interface **462** may provide, for example, for wired communication in some implementations, or for wireless communication in other implementations, and multiple interfaces may also be used.

(67) The memory **464** stores information within the mobile computing device **450**. The memory **464** can be implemented as one or more of a computer-readable medium or media, a volatile

memory unit or units, or a non-volatile memory unit or units. An expansion memory **474** may also be provided and connected to the mobile computing device **450** through an expansion interface **472**, which may include, for example, a SIMM (Single In Line Memory Module) card interface. The expansion memory **474** may provide extra storage space for the mobile computing device **450**, or may also store applications or other information for the mobile computing device **450**. Specifically, the expansion memory **474** may include instructions to carry out or supplement the processes described above, and may include secure information also. Thus, for example, the expansion memory **474** may be provided as a security module for the mobile computing device **450**, and may be programmed with instructions that permit secure use of the mobile computing device **450**. In addition, secure applications may be provided via the SIMM cards, along with additional information, such as placing identifying information on the SIMM card in a non-hackable manner.

(68) The memory may include, for example, flash memory and/or NVRAM memory (nonvolatile random access memory), as discussed below. In some implementations, instructions are stored in an information carrier such that the instructions, when executed by one or more processing devices (for example, processor **452**), perform one or more methods, such as those described above. The instructions can also be stored by one or more storage devices, such as one or more computer- or machine-readable mediums (for example, the memory **464**, the expansion memory **474**, or memory on the processor **452**). In some implementations, the instructions can be received in a propagated signal, for example, over the transceiver **468** or the external interface **462**.

(69) The mobile computing device **450** may communicate wirelessly through the communication interface **466**, which may include digital signal processing circuitry in some cases. The communication interface **466** may provide for communications under various modes or protocols, such as GSM voice calls (Global System for Mobile communications), SMS (Short Message Service), EMS (Enhanced Messaging Service), or MMS messaging (Multimedia Messaging Service), CDMA (code division multiple access), TDMA (time division multiple access), PDC (Personal Digital Cellular), WCDMA (Wideband Code Division Multiple Access), CDMA2000, or GPRS (General Packet Radio Service), LTE, 4G/6G cellular, among others. Such communication may occur, for example, through the transceiver **468** using a radio frequency. In addition, short-range communication may occur, such as using a Bluetooth, Wi-Fi, or other such transceiver (not shown). In addition, a GPS (Global Positioning System) receiver module **470** may provide additional navigation- and location-related wireless data to the mobile computing device **450**, which may be used as appropriate by applications running on the mobile computing device **450**.

(70) The mobile computing device **450** may also communicate audibly using an audio codec **460**, which may receive spoken information from a user and convert it to usable digital information. The audio codec **460** may likewise generate audible sound for a user, such as through a speaker, e.g., in a handset of the mobile computing device **450**. Such sound may include sound from voice telephone calls, may include recorded sound (e.g., voice messages, music files, among others) and may also include sound generated by applications operating on the mobile computing device **450**.

(71) The mobile computing device **450** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a cellular telephone **480**. It may also be implemented as part of a smart-phone **482**, personal digital assistant, or other similar mobile device.

(72) A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. For example, various forms of the flows shown above may be used, with steps re-ordered, added, or removed.

(73) Embodiments of the invention and all of the functional operations described in this specification can be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents,

or in combinations of one or more of them. Embodiments of the invention can be implemented as one or more computer program products, e.g., one or more modules of computer program instructions encoded on a computer readable medium for execution by, or to control the operation of, data processing apparatus. The computer readable medium can be a machine-readable storage device, a machine-readable storage substrate, a memory device, a composition of matter effecting a machine-readable propagated signal, or a combination of one or more of them. The term “data processing apparatus” encompasses all apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, or multiple processors or computers. The apparatus can include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of one or more of them. A propagated signal is an artificially generated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal that is generated to encode information for transmission to suitable receiver apparatus.

(74) A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

(75) The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit).

(76) Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Moreover, a computer can be embedded in another device, e.g., a tablet computer, a mobile telephone, a personal digital assistant (PDA), a mobile audio player, a Global Positioning System (GPS) receiver, to name just a few. Computer readable media suitable for storing computer program instructions and data include all forms of non volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

(77) To provide for interaction with a user, embodiments of the invention can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube) or LCD (liquid crystal display) monitor, for displaying information to the user and a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can

be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input.

(78) Embodiments of the invention can be implemented in a computing system that includes a back end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the invention, or any combination of one or more such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network (“LAN”) and a wide area network (“WAN”), e.g., the Internet.

(79) The computing system can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

(80) While this specification contains many specifics, these should not be construed as limitations on the scope of the invention or of what may be claimed, but rather as descriptions of features specific to particular embodiments of the invention. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

(81) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

(82) Particular embodiments of the invention have been described. Other embodiments are within the scope of the following claims. For example, the steps recited in the claims can be performed in a different order and still achieve desirable results.

Claims

1. A method comprising: obtaining a plurality of images of fish captured by a stereo camera device that comprises a pair of monocular underwater cameras; determining that one of the pair of monocular underwater cameras is at least partially obscured or is at least partially non-functional; in response to determining that one of the pair of monocular underwater cameras is at least partially obscured or is at least partially non-functional, providing the plurality of images that were captured by the other monocular underwater camera of the pair of monocular underwater cameras to a first model trained to detect one or more fish within the plurality of images from a single viewpoint; generating one or more values for each detected fish as a set of values; generating a biomass distribution of the fish based on the set of values; and determining an action based on the biomass distribution.
2. The method of claim 1, wherein generating the biomass distribution of the fish based on the set of values comprises: providing the set of values to a second model trained to estimate biomass

distributions; and obtaining output of the second model as the biomass distribution of the fish.

3. The method of claim 1, wherein the one or more values for each detected fish comprise: a value that indicates a length between a first point on a particular fish of the fish and a second point on the particular fish.

4. The method of claim 3, comprising: detecting the first point and second point on the particular fish.

5. The method of claim 1, wherein the action comprises: adjusting a feeding system providing feed to the fish.

6. The method of claim 1, wherein the action comprises: sending data including the biomass estimate to a user device, wherein the data is configured to, when displayed on the user device, present a user of the user device with a visual representation of the biomass estimate.

7. The method of claim 1, wherein the stereo camera device comprises at least two models, comprising: the first model that is trained to detect the one or more fish within the plurality of images from the single viewpoint when the one of the pair of monocular underwater cameras is at least partially obscured or is at least partially non-functional, and a stereo model that is trained to detect the one or more fish within the plurality of images from stereo images generated by the pair of monocular underwater cameras.

8. A non-transitory, computer-readable medium storing one or more instructions executable by a computer system to perform operations comprising: obtaining a plurality of images of fish captured by a stereo camera device that comprises a pair of monocular underwater cameras; determining that one of the pair of monocular underwater cameras is at least partially obscured or is at least partially non-functional; in response to determining that one of the pair of monocular underwater cameras is at least partially obscured or is at least partially non-functional, providing the plurality of images that were captured by the other monocular underwater camera of the pair of monocular underwater cameras to a first model trained to detect one or more fish within the plurality of images from a single viewpoint; generating one or more values for each detected fish as a set of values; generating a biomass distribution of the fish based on the set of values; and determining an action based on the biomass distribution.

9. The medium of claim 8, wherein generating the biomass distribution of the fish based on the set of values comprises: providing the set of values to a second model trained to estimate biomass distributions; and obtaining output of the second model as the biomass distribution of the fish.

10. The medium of claim 8, wherein the one or more values for each detected fish comprise: a value that indicates a length between a first point on a particular fish of the fish and a second point on the particular fish.

11. The medium of claim 10, wherein the operations comprise: detecting the first point and second point on the particular fish.

12. The medium of claim 8, wherein the action comprises: adjusting a feeding system providing feed to the fish.

13. The medium of claim 8, wherein the action comprises: sending data including the biomass estimate to a user device, wherein the data is configured to, when displayed on the user device, present a user of the user device with a visual representation of the biomass estimate.

14. A computer-implemented system, comprising: one or more computers; and one or more computer memory devices interoperably coupled with the one or more computers and having tangible, non-transitory, machine-readable media storing one or more instructions that, when executed by the one or more computers, perform one or more operations comprising: obtaining a plurality of images of fish captured by a stereo camera device that comprises a pair of monocular underwater cameras; determining that one of the pair of monocular underwater cameras is at least partially obscured or is at least partially non-functional; in response to determining that one of the pair of monocular underwater cameras is at least partially obscured or is at least partially non-functional, providing the plurality of images that were captured by the other monocular underwater

camera of the pair of monocular underwater cameras to a first model trained to detect one or more fish within the plurality of images from a single viewpoint; generating one or more values for each detected fish as a set of values; generating a biomass distribution of the fish based on the set of values; and determining an action based on the biomass distribution.

15. The system of claim 14, wherein generating the biomass distribution of the fish based on the set of values comprises: providing the set of values to a second model trained to estimate biomass distributions; and obtaining output of the second model as the biomass distribution of the fish.

16. The system of claim 14, wherein the one or more values for each detected fish comprise: a value that indicates a length between a first point on a particular fish of the fish and a second point on the particular fish.

17. The system of claim 16, wherein the operations comprise: detecting the first point and second point on the particular fish.

18. The system of claim 14, wherein the action comprises: adjusting a feeding system providing feed to the fish.

19. The system of claim 14, wherein the action comprises: sending data including the biomass estimate to a user device, wherein the data is configured to, when displayed on the user device, present a user of the user device with a visual representation of the biomass estimate.
